

SOCIOL 723: Social Statistics II

Week 7, Instrumental Variables

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Today

Instrumental Variable

What Does IV Estimate?

Instrument Strength

Four Canonical Applications

Applications in Sociology

Two-Stage Least Squares

Inference Under Weak Instruments

Judge and Examiner Designs

Shift-Share Instruments

Practice

★ = essential, you must understand this

★ = for understanding, not examined

A Change of Strategy

Week 6 assumed we had measured every confounder and worked on *how* to adjust. That assumption is usually false, and no diagnostic can rescue it.

The instrumental variables idea

Give up on controlling for confounders. Instead, find a source of variation in the treatment that is *as good as randomly assigned*, and use only that variation.

- We throw away most of the variation in D and keep a sliver we can defend. Precision falls; credibility rises.
- The price is steep and often hidden: we no longer estimate the ATE, and the assumptions are as untestable as ignorability was, just different, and usually easier to argue about in substantive terms.

IV

What Goes Wrong, in Scalar Form ★

We want β in $Y_i = \alpha + \beta D_i + \epsilon_i$, but D_i is **endogenous**, $\text{Cov}(D_i, \epsilon_i) \neq 0$. OLS gives

$$\begin{aligned}\hat{\beta} &= \frac{\text{Cov}_n(D_i, Y_i)}{\text{Var}_n(D_i)} = \frac{\text{Cov}_n(D_i, \alpha + \beta D_i + \epsilon_i)}{\text{Var}_n(D_i)} && \text{substitute the model for } Y_i \\ &= \beta + \frac{\text{Cov}_n(D_i, \epsilon_i)}{\text{Var}_n(D_i)} && \text{covariance is linear; the constant drops} \\ &\xrightarrow{p} \beta + \frac{\text{Cov}(D_i, \epsilon_i)}{\text{Var}(D_i)} && \text{LLN: sample moments to population moments}\end{aligned}$$

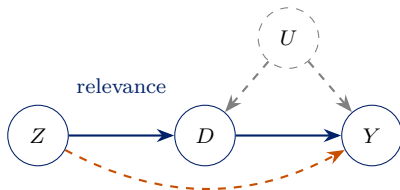
The second term does not shrink with n : OLS is biased *and* inconsistent.

Three sources of $\text{Cov}(D_i, \epsilon_i) \neq 0$: **omitted variables** (ability, family resources sit in ϵ and move with D ; Week 1's OVB formula); **simultaneity** (Y also causes D : police and crime); **selection** into the sample on a collider (Week 5).

A different problem, measurement error in D : classical error in D shrinks the OLS slope toward zero, by the reliability ratio $\text{Var}(D^*)/\text{Var}(D)$. A known direction, unlike the three above.

An Instrument ★

An instrument Z_i moves D_i but is otherwise unrelated to Y_i .



must not exist

An Instrument ★ (cont.)

Three conditions, stated carefully

1. **Relevance:** $\text{Cov}(Z_i, D_i) \neq 0$. **Testable.**
2. **Independence:** Z_i is as good as randomly assigned, no arrow into Z from anything that also causes Y .
3. **Exclusion:** Z affects Y *only* through D . **Not testable.**

Conditions 2 and 3 are distinct and are often conflated. A randomized lottery satisfies 2 by construction and can still violate 3.

The Wald Estimator ★

Take covariances of $Y_i = \alpha + \beta D_i + \epsilon_i$ with Z_i :

$$\text{Cov}(Z_i, Y_i) = \beta \text{Cov}(Z_i, D_i) + \underbrace{\text{Cov}(Z_i, \epsilon_i)}_{=0} \implies \boxed{\beta_{IV} = \frac{\text{Cov}(Z_i, Y_i)}{\text{Cov}(Z_i, D_i)}}$$

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With a *binary* instrument this is a ratio of two differences in means:

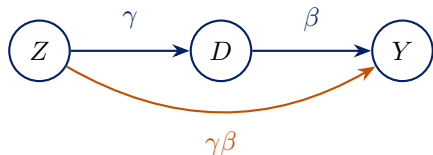
$$\beta_{IV} = \frac{E[Y_i | Z_i = 1] - E[Y_i | Z_i = 0]}{E[D_i | Z_i = 1] - E[D_i | Z_i = 0]} = \frac{\text{reduced form}}{\text{first stage}}.$$

Read it as: *the effect of the instrument on the outcome, scaled up by how much the instrument moved the treatment.*

Intuition for the Ratio ★

Divide numerator and denominator by $\text{Var}(Z_i)$:

$$\beta_{IV} = \frac{\text{Cov}(Z_i, Y_i) / \text{Var}(Z_i)}{\text{Cov}(Z_i, D_i) / \text{Var}(Z_i)} = \frac{\text{coefficient from regressing } Y \text{ on } Z}{\text{coefficient from regressing } D \text{ on } Z}.$$



- The instrument's total effect on Y travels along one path and equals $\gamma\beta$. The first stage recovers γ . Dividing isolates β .
- If γ is small, we are dividing by something close to zero. Hold that thought; it is the whole of the weak-instrument problem.

LATE

Four Types of People ★

With a binary instrument and binary treatment, write $D_i(z)$ for the treatment unit i would take if assigned $Z_i = z$. Every unit is one of four types:

Type	$D_i(0)$	$D_i(1)$	
Compliers	0	1	do what the instrument says
Always-takers	1	1	treated regardless
Never-takers	0	0	untreated regardless
Defiers	1	0	do the opposite

- We never observe anyone's type: we see $D_i(1)$ or $D_i(0)$, never both. Type is a latent variable, and this is the same fundamental problem as Week 5, one level up.

Four Types of People ★ (cont.)

- Only **compliers** have their treatment moved by the instrument.
- **Monotonicity** assumes there are no defiers: the instrument pushes everyone in the same direction, or not at all.

Deriving the LATE: the Numerator

Write the reduced form using the switching equation $Y_i = Y_i(D_i(Z_i))$ and split by type. Because Z is independent of potential outcomes and types,

$$\begin{aligned}
 & E[Y_i \mid Z_i = 1] - E[Y_i \mid Z_i = 0] \\
 &= \Pr(\text{AT}) \underbrace{(E[Y_i(1) \mid \text{AT}] - E[Y_i(1) \mid \text{AT}])}_{=0} \\
 &+ \Pr(\text{NT}) \underbrace{(E[Y_i(0) \mid \text{NT}] - E[Y_i(0) \mid \text{NT}])}_{=0} \\
 &+ \Pr(\text{C}) (E[Y_i(1) \mid \text{C}] - E[Y_i(0) \mid \text{C}]) \\
 &- \Pr(\text{D}) (E[Y_i(1) \mid \text{D}] - E[Y_i(0) \mid \text{D}]).
 \end{aligned}$$

Always- and never-takers take the *same* treatment under both values of Z , so **by exclusion** their outcomes are unchanged and they contribute zero. **Monotonicity** kills the last line by assuming $\Pr(\text{D}) = 0$, note that defiers would enter with the *wrong sign*.

Deriving the LATE: the Denominator ★

The first stage, $E[D_i | Z_i = 1] - E[D_i | Z_i = 0]$, recovers the complier share:

$$\begin{aligned} \text{first stage} &= \Pr(C) + \Pr(AT) - \Pr(AT) - \Pr(D) && \text{by type} \\ &= \Pr(C) - \Pr(D) \\ &= \Pr(C) && \text{no defiers} \end{aligned}$$

Dividing:

The LATE theorem (Imbens and Angrist 1994)

$$\frac{E[Y | Z = 1] - E[Y | Z = 0]}{E[D | Z = 1] - E[D | Z = 0]} = E[Y_i(1) - Y_i(0) | i \text{ is a complier}].$$

IV does not estimate the ATE. It estimates the average effect for the subpopulation whose treatment status the instrument actually changed, a group you cannot identify by name, defined by the instrument you happened to choose.

Living With the LATE ★

- If effects are homogeneous, $LATE = ATE$ and none of this matters. That is rarely a defensible assumption in sociology.
- Different instruments for the same treatment estimate *different* LATEs. When two IV papers disagree, that is not necessarily a contradiction; it may be two correct answers to two different questions.
- **You can describe the compliers even though you cannot name them:** their share is the first stage, and their covariate means follow because always-takers and never-takers are observed groups (Abadie's κ).
- **Compliers are often unusual:** for quarter-of-birth they are marginal dropouts; for judge-leniency designs they are defendants near the decision margin. Whether that group is the policy-relevant one is a substantive question you must answer, not a technicality.

Always characterize your compliers: without that, a LATE is an average effect for a group nobody can describe.

Describing the Compliers (Abadie 2003)

Binary Z and D , no defiers. Among $Z = 0$ the treated are always-takers; among $Z = 1$ the untreated are never-takers. So $\Pr(AT)$, $\Pr(NT)$, $E[X | AT]$, $E[X | NT]$ are observed, $\Pr(C)$ is the first stage, and the complier mean is what is left of the population mean:

$$E[X | C] = \frac{E[X] - \Pr(AT) E[X | AT] - \Pr(NT) E[X | NT]}{\Pr(C)}$$

Card: D = more than one year of college, Z = near a four-year college;
 $\Pr(AT) = 0.33$, $\Pr(NT) = 0.55$, $\Pr(C) = 0.12$ (figs/w7_card.R).

	everyone	always-takers	never-takers	compliers
share Black	0.23	0.15	0.29	0.21
share South	0.40	0.50	0.38	0.26
share metropolitan	0.71	0.58	0.78	0.77

Compliers are less Southern and more urban than always-takers. Dividing by $\Pr(C) = 0.12$ amplifies noise: rough numbers.

Can You Check Monotonicity? (1) What It Implies

Not directly, $D_i(1)$ and $D_i(0)$ are never both seen; but “no defiers” can be *refuted*.

- **One sign everywhere.** If Z only pushes D up, then $\Pr(D = 1 \mid Z = 1, X = x) \geq \Pr(D = 1 \mid Z = 0, X = x)$ in every subgroup; a negative first stage somewhere is a rejection. Card (figs/w7_card.R):

	South	non-South	metro	non-metro	Black	other
first stage	0.31	0.34	0.37	0.27	0.28	0.35

- **Balke–Pearl inequalities** (binary Z, D). Switching Z on can only *add* treated units, so for any outcome set A , $\Pr(Y \in A, D = 1 \mid Z = 1) \geq \Pr(Y \in A, D = 1 \mid Z = 0)$ and $\Pr(Y \in A, D = 0 \mid Z = 0) \geq \Pr(Y \in A, D = 0 \mid Z = 1)$. Kitagawa (2015) tests them; rejection means exclusion, independence, or monotonicity failed.
- Passing does not establish monotonicity; for lotteries and distance the case is institutional.

Can You Check Monotonicity? (2) Bounding the Damage

With defiers present and first stage $\Delta_D = \Pr(C) - \Pr(D)$:

$$\text{Wald} = \frac{\Pr(C) \tau_C - \Pr(D) \tau_D}{\Delta_D} \quad \text{numerator and denominator by type}$$

$$\tau_C = \frac{\text{Wald} \cdot \Delta_D + \Pr(D) \tau_D}{\Delta_D + \Pr(D)} \quad \text{solve for the complier effect}$$

Only $\Pr(D)$ and τ_D are unknown. Assume **at most a share δ of defiers** and a bounded outcome, $|\tau_D| \leq M$; the worst cases bound τ_C .

- **The judge toy example:** Wald = -0.20, $\Delta_D = 0.10$, $M = 1$, $\delta = 0.20$ gives $\tau_C \in [-0.73, 0.60]$: a first stage no larger than the defier share is fatal.
- **Sharp version** (Noack 2021): sensitivity parameters for the defier share and the gap $\tau_D - \tau_C$; report how large they must be before the conclusion flips.
- **Rescue without a bound** (de Chaisemartin 2017): if a subgroup of compliers matches the defiers in size and mean effect, Wald is the LATE of the remaining compliers. Untestable.

Strength

Weak Instruments Are Worse Than No Instrument ★

Recall $\hat{\beta}_{IV} = \text{Cov}_n(Z, Y) / \text{Cov}_n(Z, D)$. If the denominator is near zero, small violations of exclusion get *divided by* a small number:

$$\text{plim } \hat{\beta}_{IV} = \beta + \frac{\text{Cov}(Z, \epsilon)}{\text{Cov}(Z, D)}.$$

- An instrument that is one-tenth as endogenous as D but one-hundredth as strongly correlated with it produces **more** bias than plain OLS.
- The IV standard error is roughly the OLS standard error divided by $|\rho_{ZD}|$, the correlation between instrument and treatment: $\rho_{ZD} = 0.14$ costs a factor of 7.
- Worse, under weak identification $\hat{\beta}_{IV}$ has **no Gaussian limit**: heavy tails, so the t -test fails no matter how large n is.
- The problem *worsens* with more instruments, because many weak instruments overfit the first stage (Bound–Jaeger–Baker, in the Angrist–Krueger frames).

Diagnosing Instrument Strength ★

- **First-stage F** on the excluded instruments. The traditional rule of thumb is $F > 10$ (Staiger and Stock 1997; Stock and Yogo 2005), derived so that IV bias is under 10% of OLS bias.
- **That rule is about bias, not about test size, and it is now known to be far too lax for inference.** Lee, McCrary, Moreira, and Porter (2022) show that a conventional 5% t -test has correct size only when the first-stage F exceeds roughly **104**. Their τF procedure replaces the 1.96 critical value with a larger one read off F : at $F = 10$ the adjusted critical value is about **3.43**, so the honest interval is roughly **1.75** times wider than the printed one.
- Use a *robust* first-stage F if you use robust or clustered standard errors in the second stage.
- With several endogenous regressors, use the *Kleibergen–Paap* rank statistic; a strong F on each instrument separately is not enough.

Applications

Four Canonical Applications

Goal: read four classic instruments in one fixed order: the question, the data, the instrument, the three conditions, the main results.

	Angrist & Krueger (1991)	Card (1995)	Angrist (1990)	AJR (2001)
Unit	329,509 men born 1930–39, 1980 Census	3,010 men, NLS Young Men, 1976	white men born 1950–52, Social Security earnings records	64 former colonies, one cross-section
D	years of schooling	years of schooling	Vietnam-era military service	protection against expropriation, 0–10
Y	log weekly wage, 1979	log hourly wage, 1976	FICA earnings, 1981	log GDP per capita, 1995
Z	born in the first quarter	grew up near a four-year college	draft-eligible lottery number	log settler mortality
First stage	−0.109 (0.013)	0.337 (0.081)	0.159 (0.040)	−0.61 (0.13)

In the Literature: Angrist and Krueger (1991)

The question. What does a year of schooling (D) do to earnings (Y , log weekly wage)? Ability raises both, so OLS is biased up; reported schooling is noisy, which pulls OLS down. The two biases have unknown relative size.

The data. 1980 Census, 5% sample: 329,509 men born 1930–39 with positive earnings in 1979.

The instrument. Z = born in the first quarter of the year. Children start school in the calendar year they turn six but may leave at a fixed *age*, 16 or 17. A child born in January is older than a classmate born in December in every grade and reaches the leaving age with less schooling completed.

Angrist and Krueger: the Three Conditions

Relevance	First-quarter men have 0.109 fewer years of schooling (0.013). Testable, passes , but the first stage is a tenth of a year.
Independence	Birth quarter is unrelated to earnings potential. Threat: season of birth varies with mothers' income and education and with health at birth, which also move earnings.
Exclusion	Birth quarter affects earnings only through years of schooling. Threat: the January child is the oldest in the class; relative age may shape test scores and confidence, and earnings, directly.

- The two threats are different: the first puts an arrow *into* Z , the second an arrow *out of* Z that bypasses D .
- **Whose effect.** The men moved by the instrument are those who would have left at the minimum age. The estimate is a return to schooling at the bottom of the distribution, the quantity compulsory attendance laws care about.

Compliers in the sense of the previous section: the people whose schooling the instrument actually moved.

Angrist and Krueger: Main Results

Their Table III, men born 1930–39, first quarter against the other three:

	born Q1	born Q2–Q4	difference (se)
log weekly wage	5.892	5.903	−0.0111 (0.0027)
years of schooling	12.688	12.797	−0.109 (0.013)
Wald = $-0.0111 / -0.109$			0.102 (0.024)
OLS			0.0709 (0.0003)

- **Wald is above OLS**, not below: no sign of ability bias. Measurement error in schooling, or a high return for the compliers, can explain the ordering.
- **The price of a small first stage.** The Wald standard error is 80 times the OLS one, from the same 329,509 men. Dividing a tenth of a year into a one-percent wage gap magnifies every error in both.

Angrist and Krueger: the Cautionary Half

The paper is also the standard illustration of many weak instruments. It combines instruments, which is **2SLS**, the section after the applications, and shows the weak-identification problem at scale.

- To gain precision, the authors interacted quarter of birth with state and year of birth: up to 180 instruments.
- Bound, Jaeger, and Baker (1995): with that many weak instruments, 2SLS is biased *toward OLS*. Replacing quarter of birth with *randomly generated* instruments gave similar estimates and standard errors.
- The mechanism is overfitting the first stage: $\hat{D}$ reproduces the endogenous variation we were trying to discard.
- **More instruments is not more information.** Report the just-identified estimate beside any overidentified one.

Other instruments worth knowing: sibling sex composition for fertility, lottery admission for school quality.

In the Literature: Card (1995)

The question. The return to schooling once more: D = years of schooling, Y = log hourly wage in 1976. Same two biases as Angrist and Krueger, ability up and measurement error down.

The data. NLS Young Men: 3,010 men first interviewed in 1966, wages in 1976. The data of Lab 7 (`figs/w7_card.R`).

The instrument. Z = grew up in a county with a four-year college (68% did). A nearby college lowers the cost of attending: a student can live at home. The push is strongest for families who could not pay for room and board.

Card: the Three Conditions

Relevance	Growing up near a college adds 0.337 years of schooling (0.081), robust $F = 17.5$. Testable, passes, not strong.
Independence	Colleges sit where schools and labor markets are better, and the families who live there differ. Card conditions on family background; Lab 7 on region and metropolitan status. Argued, conditional on controls.
Exclusion	Proximity affects wages only through schooling. Threat: the same local labor market raises wages directly. The same controls carry this argument.

- Here independence and exclusion fail through the same variable, local economic conditions, so one set of controls serves both.
- **Whose effect.** Men from poorer families for whom a nearby college made attendance possible.

Card: Main Results

Lab 7's data, controls for experience, experience², Black, metropolitan, South:

	OLS	first stage	reduced form	IV = 0.0446/0.3373
estimate	0.074	0.337	0.045	0.132
(HC1 se)	(0.004)	(0.081)	(0.016)	(0.049)

- **IV is nearly twice OLS**, the opposite of ability bias. Card's reading: the compliers have a high return, and measurement error pulls OLS down.
- **Strength.** $F = 17.5$, $t = 2.72$: inside the range where the t -test misleads. Lab 7 computes the Anderson–Rubin set instead.

In the Literature: Angrist (1990)

The question. What did Vietnam-era military service (D) do to later earnings (Y)? Veterans differ from non-veterans in health, schooling, and family background, and enlistment is a choice: OLS compares the wrong men.

The data. Social Security (FICA) earnings records for white men born 1950–52, matched to their birthdates' lottery numbers; about 16,000 men per cohort. Veteran status by eligibility from the 1984 SIPP.

The instrument. $Z_i = 1$ if the man's birthdate drew a number below the year's ceiling. The 1970–72 lotteries gave every birthdate a random number; low numbers were called first, so eligible men served more often.

Angrist: the Three Conditions

Relevance	Veteran share 0.353 if eligible, 0.193 if not: first stage 0.159 (0.040). Testable, passes.
Independence	The number is a lottery draw. Holds by construction: nothing that moves earnings also moves the draw.
Exclusion	Eligibility affects earnings only through service. Threat: eligible men stayed in school to keep deferments, a path from Z to Y that bypasses service.

- The cleanest possible independence and still a live exclusion worry: the two conditions are distinct, as the DAG frame said.
- **Whose effect.** Men who served because they were drafted and would not have volunteered.

Angrist: Main Results

White men born in 1950, 1981 earnings in current dollars:

	first stage	reduced form (1981 \$)	Wald
estimate	0.159	-435.8	$-435.8/0.159 = -2,741$
(se)	(0.040)	(210.5)	about 15% of earnings

- **Reading.** Eligibility lowered earnings by \$436 on average across all eligible men; scaled by the 16 points of service it induced, service cost the compliers about \$2,700 a year.
- **Precision.** With a first stage of 0.16 the Wald standard error is about \$1,100 even with 16,000 men per cohort. A clean instrument buys credibility, not precision.

In the Literature: Acemoglu, Johnson, and Robinson (2001)

The question. Do institutions cause national development, measured as GDP per capita? D = average protection against expropriation risk 1985–95, a 0–10 rating; Y = log GDP per capita in 1995. OLS fails three ways: rich countries can afford good institutions (simultaneity); culture, geography, and history drive both (omitted variables); the rating is noisy (measurement error, pulling OLS toward zero).

The data. One cross-section of 64 former European colonies.

The instrument. Z = log mortality faced by European soldiers, bishops, and sailors in the colonies, 17th–19th centuries. Where settlers could survive, they settled and built inclusive institutions: property rights, rule of law. Where they died, colonization was extractive. Those institutional forms persisted after the disease environment stopped mattering.

AJR: the Three Conditions

Relevance	Higher settler mortality, worse institutions today: first stage -0.61 (0.13). Testable, passes.
Independence	Mortality was set by the diseases Europeans met, not by the colony's later development. Argued from history.
Exclusion	Mortality affects GDP per capita today <i>only</i> through institutions. The leading threat is geography: the diseases that killed settlers also shape agriculture, today's disease burden, and human capital, which affect output directly.

- AJR's answer to the threat is an *assumption*: geography is captured by latitude and continent dummies. A functional-form assumption doing identification work, rarely defended (Week 13).
- Later critiques questioned the mortality data itself. **The pattern:** exclusion cannot be tested, so the debate becomes one about history, measurement, and functional form. That is what an IV paper should invite.

AJR: Main Results

Their Table 4, 64 countries; coefficient on the 0–10 institutions rating:

	OLS	first stage	reduced form	2SLS
base sample	0.52	−0.61	−0.57	0.94
(se)	(0.06)	(0.13)	implied	(0.16)
with latitude	0.47	−0.51		1.00 (0.22)
with continent dummies	0.42	−0.43		0.98 (0.30)

- **IV is nearly twice OLS.** AJR read the gap as measurement error in the rating attenuating OLS. One rating point multiplies GDP per capita by $e^{0.94} = 2.6$.
- **The geography controls** that the exclusion argument leans on leave the estimate where it was; the first stage weakens and the standard error grows.

Applications in Sociology

Applications in Sociology

Goal: the same reading order on sociological questions.

	Sharkey, Torrats-Espinosa & Takyar (2017)	de Vaan & Stuart (2019)	Aksoy & Gambetta (2022)
Unit	city $\times$ year; 264 cities, 1990–2013	family $\times$ ED visit; 253,407 visits, 2010–15	province $\times$ election; 81 provinces, 1973–2018
D	community nonprofits per 100,000	source patient fills an opioid prescription	fasting hours on the median Ramadan day; religiosity
Y	log murder, violent, property crime	another household member fills one within a year	Islamist vote share; Quran-course attendance
Z	arts, medical, environmental nonprofits per 100,000	assigned ED physician's leave-one-out prescribing rate	daylength = latitude $\times$ lunar calendar
First-stage F	112	about 820 ($t = 28.6$)	daylength is the treatment

In the Literature: Sharkey, Torrats-Espinosa, and Takyar (2017)

The question. Did community nonprofits cut crime, 1990–2013? D = nonprofits per 100,000 residents focused on crime, violence, and community development; Y = log murder, violent, and property crime. OLS fails: nonprofits form where crime rises (simultaneity, pushing OLS toward zero) and where civic capacity is high (omitted variable); the IRS registry counts organizations with error.

The data. 264 cities with 100,000 or more residents, 1990–2013. Nonprofits from the IRS registry via the National Center for Charitable Statistics, crime from the Uniform Crime Reports. Long differences, and city and year fixed effects.

The instrument. Z = nonprofits per 100,000 for the arts, medical research, and the environment: organizations that share the city's funding climate and do not target violence.

Sharkey, Torrats-Espinosa, and Takyar: Conditions

Relevance	Cities where arts and medical nonprofits grow also grow community nonprofits: first stage 0.607 (0.057), $F = 112$ in long differences; 0.584 (0.049), $F = 143$ with fixed effects. Testable, strong.
Independence	A city's philanthropic climate is unrelated to its crime trend, given the controls. Threat: a philanthropic boom may signal the prosperity that cuts crime; the city-level controls carry the argument.
Exclusion	Arts, medical, and environmental nonprofits affect crime only through community nonprofits. Threat: an arts program may occupy youth or improve a block on its own. One instrument, so no overidentification test is available.

- A strong first stage settles relevance and nothing else; the other two conditions rest entirely on argument.

Sharkey, Torrats-Espinosa, and Takyar: Results

Their Tables 2–3, effect of ten more community nonprofits per 100,000 residents:

	long differences			city and year fixed effects		
	murder	violent	property	murder	violent	property
OLS	−6%	−6%	−5%	−5%	−4%	−4%
IV	−12%	−10%	−7%	−9%	−6%	−4%

- **IV exceeds OLS** in every column: measurement error in the registry attenuates OLS, or this is the effect in cities whose nonprofit sector responds to the funding climate.
- **Ordering.** Largest for murder, smallest for property crime, under both estimators and both designs.

In the Literature: de Vaan and Stuart (2019)

The question. Does opioid use spread inside the family? D = a family member, the *source patient*, fills an opioid prescription after an emergency department (ED) visit; Y = another member of the household fills one within a year. OLS fails: households share genes, partners sort on health behavior, and members share conditions and physicians.

The data. Massachusetts All Payer Claims Database, 2010–2015, every insured resident's medical and pharmacy claims; families read off shared insurance policies. Sample: 253,407 ED visits by a member of a family with no prior opioid use; 13% of the visits ended in an opioid prescription.

The instrument. Z = the opioid prescribing rate of the ED physician the source patient happened to see, computed on that physician's other visits (a leave-one-out mean over 1,075 physicians). Arriving patients get whoever is on duty.

de Vaan and Stuart: the Three Conditions

Relevance	One SD more prescribing raises the source patient's chance of a prescription from 13.0% to 15.3%: first stage 0.859 (0.030) with ED and month fixed effects and controls. Testable, very strong.
Independence	Within an ED and month, the on-duty physician is as good as randomly assigned. Balance (their Table 4): patients of high and low prescribers match on prior benzodiazepine use (.38 / .38), comorbidity index (1.08 / 1.08), sex (.50 / .50); age differs by three months.
Exclusion	The physician's habit reaches the household only through the source patient's prescription. Threat: a liberal prescriber may differ in other advice or drugs. Defense: neither patient nor family sees an ED physician again, so the visit is the only contact.

- This is an **examiner design**: the leave-one-out leniency of a randomly assigned decision-maker. A later section builds it from scratch and adds the condition it needs beyond these three.

de Vaan and Stuart: Main Results

Their Tables 5–6, family level, outcome within a year; base rate 14.6%:

	OLS, raw	OLS, FE and controls	first stage	IV
estimate	0.023	0.027	0.859	0.130
(se)	(0.002)	(0.002)	(0.030)	(0.038)
relative to base	+16%	+18%		+95%

- **IV is five times OLS.** The authors' reading: the instrument moves only households that fill what is prescribed, and the effect is large there; OLS averages in families who refuse opioids whatever the physician does. Prescriptions written but never filled are unobserved, which also pulls OLS down.
- **Robustness.** 24 OLS and 24 IV specifications (three windows, two levels, four control sets): IV effects run from +75% to +200%, every one significant.

In the Literature: Aksoy and Gambetta (2022)

The question. Does a costlier religious practice strengthen religiosity and political Islam? The cost is the length of the Ramadan fast. D = fasting hours on the median Ramadan day; Y = Islamist party vote share, and Quran-course attendance. OLS fails: provinces that fast devoutly are also poorer and more rural (omitted variables).

The data. 81 Turkish provinces $\times$ elections, 1973–2018, with province and year fixed effects; a 2014 individual survey for the mechanism step.

The instrument. Daylength on the median Ramadan day = latitude $\times$ the lunar calendar. Ramadan moves back 11 days a year, so summer Ramadans mean long fasts, more so at high latitude. The fixed effects leave only the interaction.

Aksoy and Gambetta: the Three Conditions

Relevance	Daylength <i>is</i> the length of the fast. Holds by construction.
Independence	Given province and year effects, the latitude $\times$ lunar calendar interaction is unrelated to anything else provincial. Placebo outcomes, turnout and population: null.
Exclusion	Needed only for the mechanism step, daylength $\rightarrow$ religiosity $\rightarrow$ vote. Threat: a longer fast also lowers growth (Campante and Yanagizawa-Drott 2015), a path from Z to Y not through religiosity. Their overidentification test ($\chi^2_2 = 0.00$) has the power problems of every J test.

When the instrument is the variable of interest, the reduced form is the causal estimate, and exclusion is not required for it. The IV step asks a second question, about mechanism.

Aksoy and Gambetta: Main Results

The main model is the reduced form, $Y_{it} = \mu_i + \delta_t + \alpha R_{it} + \epsilon_{it}$, R_{it} the fasting hours:

per hour of fasting	estimate	(se)
Islamist vote share, points	7.16	(2.54)
Quran-course attendance, per 1,000 residents	1.76	(0.76)
SEM total effect on AKP vote, per daylength category (2014 survey)	0.03	(0.009)

- **Reading.** A half-hour longer fast adds 3.5 points of vote share, 11% of the 31% mean.
- **The IV step** instruments latent religiosity with daylength in the survey; it says the vote moves *because* religiosity moves, if exclusion holds.

2SLS

The Two Stages ★

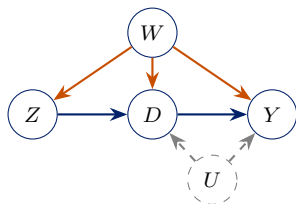
With covariates and possibly several instruments:

$$\text{Stage 1: } D_i = \pi_0 + \pi_1 Z_i + \pi_2' W_i + v_i \implies \hat{D}_i$$

$$\text{Stage 2: } Y_i = \alpha + \beta \hat{D}_i + \gamma' W_i + u_i$$

- $\hat{D}_i$ is the part of D_i *explained by the instrument*, which by assumption is uncorrelated with ϵ_i . We regress on the clean part and discard the rest.
- The exogenous covariates W_i must appear in *both* stages. Omitting them from the first stage is a common and serious error.
- **Order condition:** at least as many excluded instruments as endogenous regressors. *Exactly identified* means equality; *overidentified* means more.
- Never run the two stages by hand. The second-stage standard errors would treat $\hat{D}$ as data rather than as an estimate. Use `ivreg()` or `feols(y ~ w | d ~ z)`.

Why W Goes in Both Stages



$Z \leftarrow W \rightarrow Y$ is a back door from instrument to outcome: Z is valid only *given* W . 2SLS with W in both stages partials W out of Y , D , Z , then takes the one- Z Wald ratio (FWL). W belongs in the first stage when it affects Z , in the second when it affects Y .

Simulation, true $\beta = 1$, $n = 5,000$, 500 draws (figs/w7_twostage_W.R); mean $\hat{\beta}$:

W enters	$W \rightarrow Z$	$Z \perp W$
both stages	1.00	1.00
second stage only	0.56	1.00
first stage only	1.71	1.90
neither	1.55	1.00

- **Second stage only.** $\hat{D}$ from Z alone carries W 's variation, and $D - \hat{D}$ is correlated with W in the second-stage error. Harmless only if $Z \perp W$ (a lottery).
- **First stage only.** $\hat{D}$ contains W , and $W \rightarrow Y$ is in the error: exclusion fails for $\hat{D}$.

IV Standard Error, Scalar Version (1)

One Y , one D , one Z . Substitute the model for Y_i into the Wald ratio:

$$\begin{aligned}\hat{\beta}_{IV} &= \frac{\text{Cov}_n(Z_i, Y_i)}{\text{Cov}_n(Z_i, D_i)} = \frac{\text{Cov}_n(Z_i, \alpha + \beta D_i + \epsilon_i)}{\text{Cov}_n(Z_i, D_i)} && \text{substitute } Y_i = \alpha + \beta D_i + \epsilon_i \\ &= \beta + \frac{\text{Cov}_n(Z_i, \epsilon_i)}{\text{Cov}_n(Z_i, D_i)} && \text{covariance is linear; the constant drops} \\ &= \beta + \frac{\sum_i \tilde{z}_i \epsilon_i}{\sum_i \tilde{z}_i \tilde{d}_i} && \text{centered } Z \text{ and } D; \text{ the } 1/n \text{ cancel}\end{aligned}$$

- **The numerator is the random part:** independent terms $\tilde{z}_i \epsilon_i$ with mean zero and variance $\tilde{z}_i^2 \sigma_i^2$; its variance is their sum.
- **The denominator is treated as fixed** (condition on Z and D): the “strong instrument” step; the weak-IV section is about its failure.

$$\text{Var}(\hat{\beta}_{IV} \mid Z, D) = \frac{\sum_i \tilde{z}_i^2 \sigma_i^2}{(\sum_i \tilde{z}_i \tilde{d}_i)^2}.$$

IV Standard Error, Scalar Version (2)

Take $\sigma_i^2 = \sigma^2$ for every i and divide top and bottom by $\sum_i \tilde{z}_i^2 \sum_i \tilde{d}_i^2$; ρ_{ZD} is the correlation of Z and D :

$$\text{Var}(\hat{\beta}_{IV}) = \frac{\sigma^2 \sum_i \tilde{z}_i^2}{(\sum_i \tilde{z}_i \tilde{d}_i)^2} = \frac{\sigma^2}{\sum_i \tilde{d}_i^2} \cdot \frac{\sum_i \tilde{z}_i^2 \sum_i \tilde{d}_i^2}{(\sum_i \tilde{z}_i \tilde{d}_i)^2} = \underbrace{\frac{\sigma^2}{n s_D^2}}_{\text{OLS variance}} \cdot \frac{1}{\rho_{ZD}^2}$$

- **IV standard error = OLS standard error / $|\rho_{ZD}|$** , for the same σ . The instrument explains a fraction ρ_{ZD}^2 of the variance in D , and only that fraction is used: $\rho_{ZD} = 0.14$ costs a factor of 7.
- **Which residual.** σ^2 is estimated from $\hat{\epsilon}_i = Y_i - \hat{\alpha} - \hat{\beta}_{IV} D_i$, with the *actual* D_i . A second stage run by hand uses $\hat{D}_i$ instead, gets a smaller residual, and prints too small a standard error.
- **Robust version.** Keep σ_i^2 and plug in $\hat{\epsilon}_i^2$: $\widehat{\text{Var}} = \sum_i \tilde{z}_i^2 \hat{\epsilon}_i^2 / (\sum_i \tilde{z}_i \tilde{d}_i)^2$.

The Scalar Formula on Card's Data

Log wage on schooling, instrument near a four-year college, no controls ($n = 3,010$)
(figs/w7_card.R).

	estimate	standard error
OLS	0.052	0.00287
IV, homoskedastic formula	0.188	$0.00287 \times 1.321 / 0.1442 = 0.0263$
IV, robust formula	0.188	0.0261 (ivreg + vcovHC: 0.0261)
second stage by hand	0.188	0.0207 too small

- $\rho_{ZD} = 0.144$: proximity explains 2% of the variance in schooling, so $1/|\rho_{ZD}| = 6.9$.
- The IV residual is larger than the OLS residual (sd 0.557 against 0.421) because the IV slope is not the least-squares slope: another factor of 1.32. Together, 9.2 times the OLS standard error.
- By hand, the second-stage residual has sd 0.438, not 0.557: 21% too small.

IV in Matrix Form ★

Start from the moment condition $E[\mathbf{Z}'u] = \mathbf{0}$ with $Y = D\beta_0 + u$. The sample analogue $\mathbf{Z}'(\mathbf{Y} - \mathbf{D}\hat{\beta}) = \mathbf{0}$ has no solution when there are more instruments than regressors, so instead *minimize* the sample moments in a weighted norm:

$$\hat{\beta} = \arg \min_b (\mathbf{Z}'(\mathbf{Y} - \mathbf{D}b))'(\mathbf{Z}'\mathbf{Z})^{-1}(\mathbf{Z}'(\mathbf{Y} - \mathbf{D}b)).$$

The first-order condition gives

$$\hat{\beta} = \left(\mathbf{D}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{D} \right)^{-1} \mathbf{D}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{Y}.$$

IV in Matrix Form: The 2SLS Estimator ★

Define the projection matrix $\mathbf{P}_Z = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'$, the same object as Week 1, now projecting onto the instrument space:

$$\hat{\beta}_{2SLS} = (\mathbf{D}'\mathbf{P}_Z\mathbf{D})^{-1}\mathbf{D}'\mathbf{P}_Z\mathbf{Y}$$

Since $\mathbf{P}_Z\mathbf{D}$ is the first-stage fitted value and $\mathbf{P}_Z$ is idempotent, this is literally “regress $\mathbf{Y}$ on $\hat{\mathbf{D}}$.” And when $\mathbf{Z} = \mathbf{D}$, $\mathbf{P}_Z\mathbf{D} = \mathbf{D}$ and 2SLS collapses to OLS: OLS is the special case in which you claim everything is exogenous.

The IV Sandwich ★

Substitute $\mathbf{Y} = \mathbf{D}\beta_0 + \mathbf{u}$:

$$\hat{\beta} - \beta_0 = (\mathbf{D}'\mathbf{P}_Z\mathbf{D})^{-1}\mathbf{D}'\mathbf{P}_Z\mathbf{u}.$$

Multiply by $\sqrt{n}$. Under regularity conditions and a *strong* instrument,

$$\frac{1}{n}\mathbf{D}'\mathbf{P}_Z\mathbf{D} \xrightarrow{p} \mathbf{A} \succ 0, \quad \frac{1}{\sqrt{n}}\mathbf{D}'\mathbf{P}_Z\mathbf{u} \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{S}),$$

so by Slutsky $\sqrt{n}(\hat{\beta} - \beta_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{A}^{-1}\mathbf{S}\mathbf{A}^{-1})$.

The robust variance estimator is the familiar sandwich:

$$\hat{V}(\hat{\beta}) = (\mathbf{D}'\mathbf{P}_Z\mathbf{D})^{-1}\mathbf{D}'\mathbf{P}_Z\hat{V}(\mathbf{u})\mathbf{P}_Z\mathbf{D}(\mathbf{D}'\mathbf{P}_Z\mathbf{D})^{-1},$$

plugging in $\hat{V}(\mathbf{u}) = \text{diag}(\hat{u}_i^2)$, and clustering exactly as in Week 2. **Note the residuals $\hat{u}_i$ must be computed with the actual D_i , not $\hat{D}_i$** ; this is the other reason not to run the stages by hand.

Overidentification Tests: Read Them Carefully

With more instruments than endogenous regressors, the Sargan–Hansen J test asks whether the instruments agree with one another.

- **Rejection** says at least one instrument is invalid, *or* that effects are heterogeneous, so the instruments estimate different LATEs and are supposed to disagree. It does not say which.
- **Non-rejection is nearly worthless.** It is consistent with all the instruments being invalid *in the same direction*, and the test has little power.
- The J test cannot validate exclusion. Nothing can. The defence is substantive: tell a story about why Z could not affect Y except through D , and look for *placebo outcomes* it should not move and *placebo samples* in which the first stage should be absent.

Weak IV

Where $E[(Y - \beta D) Z] = 0$ Comes From

The IV assumption: the instrument is unrelated to the error, $\text{Cov}(Z_i, \epsilon_i) = 0$.

Substitute what the error is:

$$\epsilon_i = Y_i - \alpha - \beta D_i \quad \text{the structural equation, solved for } \epsilon_i$$

$$0 = \text{Cov}(Z_i, \epsilon_i) = \text{Cov}(Z_i, Y_i - \beta D_i) \quad \alpha \text{ is a constant}$$

$$= \text{Cov}(Z_i, Y_i) - \beta \text{Cov}(Z_i, D_i) \quad \text{covariance is linear}$$

- **In words:** strip the treatment's part, βD , out of Y ; what remains must be unrelated to Z . The true β satisfies this equation.
- **Solve it for β** and you get the Wald estimator, $\text{Cov}(Z, Y) / \text{Cov}(Z, D)$. Nothing new is assumed.
- $E[(Y_i - \beta D_i) \tilde{Z}_i]$ with $\tilde{Z}_i = Z_i - E[Z_i]$ is the same covariance; the tilde only centers Z .

Two Ways to Use the Same Equation

Solve it (Wald, 2SLS):

$\hat{\beta}_{IV} = \text{Cov}_n(Z, Y) / \text{Cov}_n(Z, D)$. Divides by the first stage; unreliable when that is small and noisy.

Check it (Anderson–Rubin): guess β_0 , ask whether $\text{Cov}_n(Z, Y - \beta_0 D) = 0$. Never divides.

Card's data (Z = near a college, D = schooling, Y = log wage):

$\text{Cov}_n(Z, Y) = 0.0338$, $\text{Cov}_n(Z, D) = 0.1798$ (figs/w7_weakiv.R).

guess β_0	0	0.10	0.144	0.188	0.25	0.30
$\text{Cov}_n(Z, Y - \beta_0 D) = 0.0338 - \beta_0 \times 0.1798$	0.034	0.016	0.008	0.000	-0.011	-0.020

Zero at $\beta_0 = 0.188$, the Wald estimate. For the other guesses: how far from zero is too far? That needs a scale.

The AR Statistic, Step by Step

1. Guess β_0 ; build $U_i = Y_i - \beta_0 D_i$.
2. Regress U_i on Z_i . Since $U_i = \alpha + (\beta - \beta_0)D_i + \epsilon_i$, the true slope is $(\beta - \beta_0)\pi$: zero if the guess is right.
3. $C(\beta_0) = t^2$ of that slope; χ_1^2 under $H_0: \beta = \beta_0$. Reject the guess if $C(\beta_0) > 3.84$.

guess β_0	0	0.10	0.144	0.188	0.25	0.30
slope of U on Z	0.156	0.073	0.037	0.000	-0.051	-0.093
t	9.10	4.25	1.91	0.00	-1.94	-3.02
$C(\beta_0) = t^2$	82.7	18.0	3.66	0.00	3.76	9.12
verdict	reject	reject	keep	keep	keep	reject

Step 2 is an ordinary regression t -test: $\hat{\pi}$ never appears in a denominator. A weak instrument shrinks $(\beta - \beta_0)\pi$ for every guess, so fewer guesses are rejected: a wider set, not a wrong one.

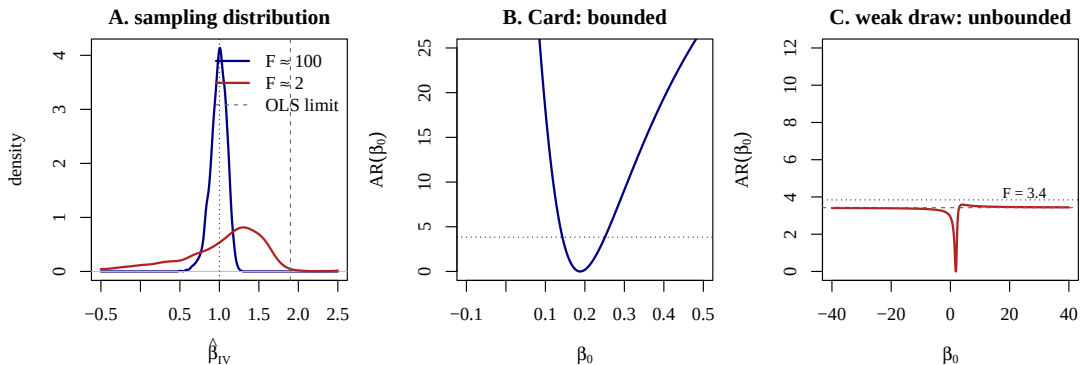
Inverting the Test: the AR Confidence Region

Keep every guess the data cannot reject, $CR(\beta) = \{\beta_0 : C(\beta_0) \leq c_{1-\alpha}\}$. “**Works under a weak instrument**” means: covers the truth 95% of the time whatever F is. Simulation, $n = 500$, $\beta = 1$, 4,000 draws (figs/w7_weakiv.R):

median first-stage F	1.8	6.0	9.7	101
Wald interval covers the truth	87%	91%	92%	95%
AR region covers the truth	94%	95%	95%	95%
AR region is a bounded interval	26%	69%	87%	100%
median width, Wald	2.76	1.59	1.26	0.39
median width, AR when bounded	2.29	2.15	1.70	0.40

- **Why Wald fails:** its standard error is too *small*, not too large; $\hat{\beta}_{IV}$ leans toward OLS with heavy tails, and the printed normal interval is not there.
- **What AR does:** at $F \approx 6$, a bounded interval in 69% of draws, wider than Wald's, else an unbounded region: it admits what the data cannot pin down, and coverage stays at 95%.

The AR Statistic on Card's Data and on the Weak Draw

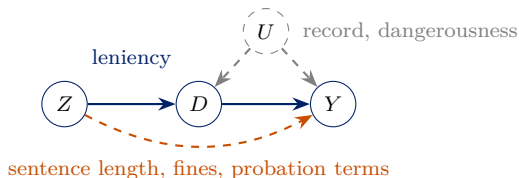


A: sampling distribution of $\hat{\beta}_{IV}$ with a strong and a weak instrument, 4,000 draws each. B and C: $C(\beta_0)$ against β_0 ; the dotted line is 3.84 and the dashed line the first-stage F . The AR region is where the curve lies below 3.84: an interval for Card, the whole line for the weak draw (figs/w7_weakiv.R).

Judge IV

The Design

- **Goal:** among people *convicted of a felony*, the effect of a prison sentence ($D = 1$) rather than probation ($D = 0$) on later reoffending or employment Y . The pool is the convicted, not the public.
- **Problem:** within that pool, judges send the worse records and graver offenses to prison, and the same things predict Y ; prisoners and probationers are not comparable.
- **Variation:** cases are assigned to judges at random within a court and year, and judges differ persistently in how often they imprison. Z_i = the leniency of the judge defendant i drew.



Constructing the Instrument

Leave-one-out leniency: your judge's incarceration rate in everyone else's cases (within court and year), $Z_i = \frac{1}{n_{j(i)}-1} \sum_{k \neq i: j(k)=j(i)} D_k$.

- **Why leave your own case out.** The leave-in mean contains D_i , which is correlated with ϵ_i ; the instrument inherits $\text{Cov}(D_i, \epsilon_i)/n_j$ of it, shrinking with cases per judge only.
- **Judge dummies are the leave-in mean in disguise.** With $J - 1$ judge indicators as instruments (Harding et al.; Kling 2006) the fitted value is the judge's rate over *all* cases, own case included: same bias.

Simulation (figs/w7_judge.R): 100 judges, a confounder raising D and Y , true $\tau = 0.20$, 500 replications.

cases per judge	OLS	judge dummies = leave-in (identical every time)	leave-one-out
50	1.17	0.56	0.17 (sd 0.29)
200	1.17	0.33	0.20 (sd 0.12)

The Three Conditions, Judge Version

- **Relevance.** Judges differ and courts have thousands of cases; F is rarely the problem. Compute Z and F within court $\times$ year.
- **Independence.** Random assignment is an institutional fact and partly testable: regress Z on defendant and case characteristics (race, age, priors, offense); nothing should predict the draw. Michigan assigns by a random number generator (Harding et al. 2019).
- **Exclusion.** The harsh judge must change Y only through prison versus probation. The same judge also sets sentence length, fines, and probation conditions, and speaks differently from the bench: each is a path from Z to Y . Remedies: instrument every dimension of the sentence at once (Harding et al.: prison, jail, jail with probation, three sentence lengths), or define the treatment as the whole sentence package.
- **Monotonicity.** With two judges it reads “no defiers.” With many judges it must hold for every pair of judges and every defendant.

Monotonicity With Many Judges: a Worked Failure

Two judges, two case types in equal shares.

share incarcerated	drug cases (50%)	property cases (50%)	all cases
Judge A	0.90	0.30	0.60
Judge B	0.50	0.50	0.50

A is harsher overall, so $Z = 1$ for A. Among drug cases A creates 0.40 compliers; among property cases A is the lenient one and creates 0.20 **defiers**.

$$\text{first stage} = 0.5(0.40) - 0.5(0.20) = 0.10 \quad \text{defiers subtract}$$

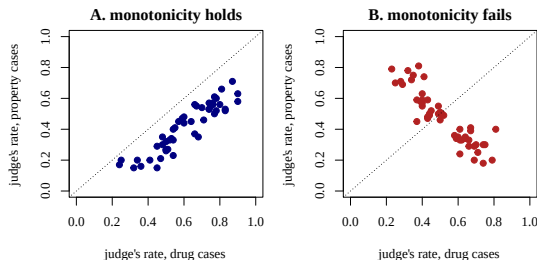
$$\text{reduced form} = 0.5(0.40) \tau_{\text{drug}} - 0.5(0.20) \tau_{\text{prop}} \quad \text{numerator of the LATE derivation}$$

$$\text{Wald} = 2 \tau_{\text{drug}} - \tau_{\text{prop}} \quad \text{weights sum to one; one is negative}$$

With $\tau_{\text{drug}} = 0.10$ and $\tau_{\text{prop}} = 0.40$ the Wald estimand is -0.20 : every effect is positive, the IV estimand is negative. Within each case type the ordering is stable and the Wald ratio recovers that type's τ .

Detecting a Monotonicity Failure

If judge A is harsher than B for one kind of defendant, A must be harsher for every kind: plot each judge's rate on one case type against the other (Dobbie, Goldin, and Yang 2018; `figs/w7_mono.R`).



- **Holds:** judges line up alike on both types; correlation of the two rates 0.92, 88% of judge pairs ordered the same way.
- **Fails:** harsh on drugs means lenient on property; correlation -0.90 , 15% of pairs alike. The pooled first stage still looks strong.

Addressing a Monotonicity Failure

- **The common fix: build leniency within strata.** Compute Z_i from the judge's other cases of the *same type* as case i (case type $\times$ court $\times$ year), and let the first stage differ by stratum. Within a stratum judges are consistently ordered, the Wald ratio recovers that stratum's τ , and 2SLS pools the strata with positive weights.
- **The common assumption: average monotonicity** (Mogstad, Torgovitsky, and Walters 2021). 2SLS with a leniency instrument is a positively weighted average of effects if, for every defendant, leniency and that defendant's own treatment covary non-negatively across judges. Judges may disagree on some cases; no defendant's treatment may run against leniency on average. Plausible when judge differences are mostly a common harshness, as in panel A.
- **A formal test** (Frandsen, Lefgren, and Leslie 2023): with a bounded outcome, exclusion and monotonicity bound the slope of the per-judge mean outcome against the per-judge incarceration rate by the range of Y . `ivcheck::iv_testjfe` in R.

Whose Effect?

- **Compliers are the marginal defendants:** those whose sentence depends on which judge they drew. Not the mandatory minimums, not first-degree murder, not the obvious probation cases. In the toy example the pairwise complier share is the first stage, 0.10.
- With a continuous Z , 2SLS averages pairwise LATEs across the leniency distribution, more weight where judges are dense.
- **Describe them.** Complier means of covariates are recoverable by Abadie's κ reweighting; judge papers report a table of complier characteristics next to the full sample.
- **For once the LATE is the policy parameter.** A sentencing reform moves exactly the marginal cases, not the always-incarcerated.
- **Different margins, different compliers.** Prison versus probation at sentencing (Harding et al. 2017) and detention versus release before trial (Dobbie, Goldin, and Yang 2018) are different populations; their LATEs need not agree.

In the Literature: Harding, Morenoff, Nguyen, and Bushway

Data: everyone sentenced for a felony in Michigan, 2003–2006 ($n = 111,110$), administrative records to 2013–2015; cases with no judicial discretion dropped.

Instruments: judge dummies and judge $\times$ presentence-characteristic interactions in 2SLS, six first stages; LIML as a check.

- **PNAS 2017.** Prison rather than probation raises the probability of being back in prison within five years by about 20 points for nonwhite and white defendants alike; technical parole violations account for 18 and 14 of those points, new felony sentences for 4 and 9. No effect on new felony convictions after release.
- **AJS 2018.** The employment effect runs mainly through incapacitation, the sentence and the later returns; post-release penalties are concentrated among whites with a presentence work history.
- **Nature Human Behaviour 2019.** Violent arrests fall by 8.4 points over five years when time inside is counted; after release, no detectable effect on violence.

Reading Harding et al. with This Week's Tools

- **Many instruments.** Judge dummies times characteristics run to hundreds; 2SLS leans toward OLS, hence the LIML check. One leave-one-out leniency per endogenous variable is now the default (Dobbie, Goldin, and Yang 2018; Bhuller et al. 2020).
- **Exclusion** is handled by widening the treatment: six sentence dimensions instrumented together.
- **Monotonicity** is assumed, not tested.
- **The estimand** is the LATE for marginal felony defendants, the population a sentencing reform would move.

Same design elsewhere. Green and Winik (2010): 1,003 D.C. drug defendants on nine random calendars, no effect on re-arrest. Loeffler (2013): 20,000 Cook County felony defendants, no effects on recidivism or employment. In economics, Kling (2006), Dobbie, Goldin, and Yang (2018), and Bhuller et al. (2020), where Norwegian prisons reduce reoffending. Guide: Chyn, Frandsen, and Leslie (2025).

Shift-share

The Problem and the Instrument

Goal: the effect of a local shock x_ℓ (job loss, immigration, import competition) on a local outcome y_ℓ (marriage, mortality, votes) when x_ℓ responds to local conditions that also move y_ℓ .

Idea (Bartik 1991; Card 2001): predict the local shock from national shocks by industry, weighted by baseline exposure,

$$z_\ell = \sum_{k=1}^K s_{\ell k} g_k,$$

$s_{\ell k}$ the baseline share of industry k in place ℓ , g_k the national change in k . Card (2001): past settlement shares by origin country, times national inflows by origin.

Plan

1. Three places and two industries, by hand.
2. Two routes to validity: exogenous shares or exogenous shocks.
3. Proof by simulation, then inference.
4. Autor, Dorn, and Hanson (2019) on marriage.

Three Places, Two Industries, by Hand

National changes over the decade: manufacturing $g_M = -10\%$, services $g_S = +5\%$.
Baseline employment shares:

place	$s_{\ell M}$	$s_{\ell S}$	$z_\ell = s_{\ell M}g_M + s_{\ell S}g_S$
A	0.6	0.4	$0.6(-10) + 0.4(5) = -4$
B	0.2	0.8	$0.2(-10) + 0.8(5) = +2$
C	0.4	0.6	$0.4(-10) + 0.6(5) = -1$

- The shocks are common to every place; z_ℓ varies only through the shares. The design compares A with B: the place that was manufacturing when manufacturing collapsed.
- Estimation is 2SLS of y_ℓ on x_ℓ with z_ℓ as the one excluded instrument, controls in both stages. K industries are not K instruments; Bound–Jaeger–Baker does not apply. What K changes is the inference.

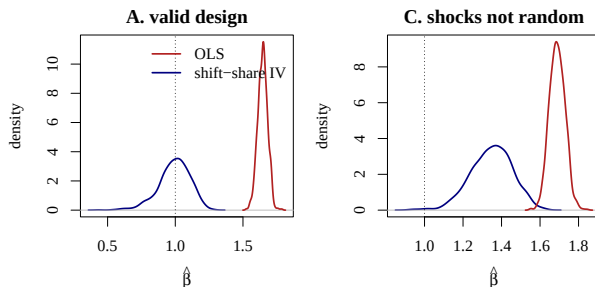
Two Routes to Validity

	shares exogenous (Goldsmith-Pinkham, Sorkin, and Swift 2020)	shocks exogenous (Borusyak, Hull, and Jaravel 2022)
as good as random	the baseline shares $s_{\ell k}$, given controls	the shocks g_k across industries; shares may be endogenous
the other piece	shocks only set the weights; the instrument is really the K share variables	shares only set the exposure; the shocks are the identifying variation
reads like	a difference-in-differences with continuous exposure: high-share places would have trended like low-share ones	an industry-level IV with K observations weighted by average exposure $\bar{s}_k$
typical setting	Card enclaves; few, large shocks	the China shock; many industries, idiosyncratic shocks
what to check	pre-trends by exposure; Rotemberg weights, which shares carry the estimate	shock balance, do the g_k predict industry averages of covariates; effective number of shocks $1/\sum_k \bar{s}_k^2$

Choose the story that fits the setting and defend that one; the two require different checks.

Proof by Simulation

300 places, 80 industries, true $\beta = 1$, 1,000 replications, first-stage $R^2 \approx 0.20$. A local supply shock enters both x_ℓ and y_ℓ , so OLS is biased (figs/w7_shiftshare.R).



scenario	OLS	SSIV
A. both stories hold	1.65	0.99
B. shares endogenous	1.65	0.98
C. shocks not random	1.69	1.35

- **B** (shocks random) is the Borusyak–Hull–Jaravel point: an endogenous share does no harm, because z_ℓ averages many mean-zero shocks.
- **C** is the failure neither story survives: a national shock that moves y through a channel other than x (the industry's technology, unions, politics).

Inference: Places That Look Alike Are Not Independent

Adão, Kolesár, and Morales (2019): if y_ℓ carries an industry component $\sum_k s_{\ell k} \nu_k$, places with similar shares have correlated residuals wherever they sit on the map. Robust and state-clustered standard errors miss this and over-reject. Coverage of 95% intervals in the simulation (figs/w7_shiftshare.R):

	robust SE	shock-level SE
D. valid, industry-correlated errors	0.90	0.94
C. shocks not random (biased)	0.07	0.13

- **Shock-level regression** (Borusyak, Hull, and Jaravel 2022): average y_ℓ and x_ℓ to industries with exposure weights, instrument with g_k , weight by $\bar{s}_k$. Same point estimate as the place-level 2SLS; the standard error treats the K shocks as the independent units (`ShiftShareSE` implements Adão et al.).
- Few effective shocks behave like a small sample: report $1/\sum_k \bar{s}_k^2$ next to the first-stage F .

In the Literature: Autor, Dorn, and Hanson (2019)

The title is borrowed from Wilson (1996): does the collapse of manufacturing lower young men's value in the marriage market?

- **Shock:** growth in Chinese import penetration per worker in commuting zone ℓ , $x_\ell = \sum_k s_{\ell k} \Delta M_k / L_k$, with male and female employment shares separately.
- **Instrument:** the same sum with China's export growth to eight other high-income countries and lagged shares (Autor, Dorn, and Hanson 2013): the exclusion claim is about shocks, Chinese productivity rather than U.S. demand. 722 commuting zones, 1990–2000 and 2000–2014 stacked, standard errors clustered by state.
- **Results:** a unit shock, the average decade-level rise in exposure, lowers the share of women aged 18–39 currently married by 0.95 points and raises the never-married share by 1.2. Shocks to male-intensive industries do the work.

Shift-share designs are only now arriving in sociology journals; the first China-shock paper in one is next.

In the Literature: Shen (2026)

Shen (2026), *Sociological Science*: did the deterioration of men's labor markets raise the correlation between spouses' earnings? The usual story credits wives' rising employment; this one asks about husbands' decline.

- **Data.** Census and ACS, 722 commuting zones, two stacked decadal changes (1990–2000, 2000–2014), $n = 1,444$. Outcome: the change in the zone's spousal earnings correlation.
- **Two shocks.** The Autor–Dorn–Hanson exposure $\sum_j (L_{ij}/L_i) \Delta Z_{ij}^u$ is split by each industry's male and female employment share in the zone, giving a male-specific and a female-specific import shock (mining versus textiles).
- **Two instruments.** The same sums built from China's export growth to eight other high-income countries and 1980 industry shares. Two endogenous regressors, so strength is reported as the Sanderson–Windmeijer conditional F (113 and 124) and the Kleibergen–Paap F (62). Standard errors clustered by state.

Shen (2026): Results and Reading

change in spousal earnings correlation	all couples	with children	without
male-specific shock	0.04 (0.02)	0.06 (0.02)	0.01 (0.02)
female-specific shock	-0.01 (0.01)	-0.02 (0.01)	-0.01 (0.02)

- **Mechanism.** A unit male shock adds 121 (33) annual work hours to mothers married to top-tercile fathers, 79 (23) in the middle, 12 (27) at the bottom: families adapt to men's decline unequally, and the adaptation itself widens inequality between households.
- **Exclusion is the shocks story:** the non-U.S. component of Chinese export growth is as good as random with respect to local determinants of the correlation, given controls (Borusyak, Hull, and Jaravel 2022).
- **Two threats the paper names.** Women are scarce in tradable manufacturing, so the female shock measures women's conditions poorly; and trade shocks also change who marries (Autor, Dorn, and Hanson 2019).

Practice

A Reader's Checklist ★

Adapted from Sovey and Green (2011). Any IV paper should report:

1. The instrument, and an explicit argument for exclusion, not just a citation to a prior paper that used it.
2. The **first stage**, its coefficient, and its robust F .
3. The **reduced form** (Y on Z), with its estimate and interval. It needs no instrument-strength assumption, so it is the most robust piece of evidence in the design, and if it is indistinguishable from zero the IV estimate rests on very little.
4. The **OLS** estimate alongside, so the reader can see the direction and size of the correction, and whether it goes the way theory predicts.
5. Who the **compliers** are, and their share.
6. Weak-instrument-robust inference if F is not comfortably large.
7. The just-identified estimate if the paper is overidentified.

A Reader's Checklist ★ (cont.)

8. Placebo checks: outcomes and subgroups Z should not affect.

If a paper reports only the 2SLS coefficient, that is a red flag.

When IV Is the Wrong Tool ★

- **No credible instrument exists.** This is the normal state of affairs. An implausible instrument is worse than an honest selection-on-observables analysis with sensitivity bounds.
- **The LATE is not the quantity of interest.** If your policy question concerns always-takers, the compliers' effect does not answer it.
- **Interpretation drifts.** Papers routinely estimate a LATE and then discuss it as if it were an ATE.
- **Monotonicity is doing hidden work.** In judge-leniency and examiner designs, it must hold for *every* judge simultaneously, a much stronger claim than it first appears.

When IV Is the Wrong Tool ★ (cont.)

The design-based cousins

When the instrument is a discontinuity in an assignment rule, you have **RDD** (Week 9), and indeed fuzzy RD *is* local IV. When it is a policy change over time, you have **DID** (Week 10).

Looking Ahead

- IV replaces “I measured all confounders” with “I found variation that is as good as random.” Both are assumptions; the second is easier to argue about substantively, and easier for readers to attack, which is a virtue.
- The price is the estimand: LATE, for compliers you cannot name, defined by the instrument you chose.
- Notice how much identification work was done by *functional form* in AJR, “linear in distance from the equator, plus continent dummies.” Week 13 asks briefly what can be done when you are unwilling to assume that.

Midterm

Thursday, October 15, in class, covering Weeks 1–7. Open book and open note, timed, no coding. Tuesday October 13 is Fall break.

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